INCIDENT RESOLUTION WORKFLOW

Level 2

AAV2011IFA01_

Ingeteam Power Technology S.A.

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1. LIST OF ALARMS

Code	Alarm	Reason for shutdown	Description
0001H	Grid frequency out of range	Grid frequency out of range	The measured grid frequency is outside the specified limits
0002H	Grid voltage out of range	Grid voltage out of range	The measured grid voltage is outside the specified limits
0004H	Control current saturation	Control current saturation	Unit control current saturation
0008H	Reset	Reset	Indicates that the inverter has reset, inverter firmware fault
0010H	Effective grid current	Effective grid current	RMS current exceeds maximum allowed
0020H	Temperature	Power electronics temperature	The temperature of power electronics exceeds 80 ºC.
		Auxiliary temperature sensor	The auxiliary temperature sensor has triggered an alarm
0040H	Hardware fault	Error reading ADC	A higher than normal reading of the top ADC has been detected on an unexpected input. Synchronism or auxiliary power supply has been lost.
		ADC latency	Internal error of the analogue to digital converter
		Master-slave	Error on master-slave system or optical fiber
		DC contactor	The status of the DC contactor is not correct for the current inverter status
		Fault in LVRT kit	Error in the optional hardware necessary to comply with the regulations on withstanding grid voltage drop outs
		Synchronization	
0080H	Instantaneous grid current	Instantaneous grid current	Instantaneous current value out of range
0100H	AC protection	Varistors	Error in AC varistors
		Contactor	The state of the contactor is not correct for the current inverter status.
		AC protection	Error in some of the AC protectors, arresters, fuses, etc.
0200H	DC protection	DC protection	Blown DC input fuses or DC arresters
0400H	DC insulation	DC insulation	Insulation fault in the PV array on inside the inverter
0800H	Branch fault alarm	Fault in branch no. 1	Fault in branch no. 1 of the power electronics
		Fault in branch no. 2	Fault in branch no. 2 of the power electronics
		Fault in branch no. 3	Fault in branch no. 3 of the power electronics
1000H	Manual shutdown	Manual shutdown	Manual shutdown due to the emergency push-button, the display or communications or a badly closed door on the unit.
2000H	Configuration	Configuration	Shutdown due to a change in firmware
		Firmware	Shutdown due to firmware load
4000H	High input voltage	High input voltage	High DC input voltage
8000H	Panel voltage	Low input voltage	Shutdown due to low input voltage
		Hardware fault	Reason for internal shutdown, hardware fault
		Low power	Shutdown due to low power, (typical at nightfall)
7FFFH	Fatal error	Fatal error	Due to five consecutive branch fault errors, consecutive contactor shutdowns, an error in any of the fans or high temperature fault
	Com Error	Communications error	

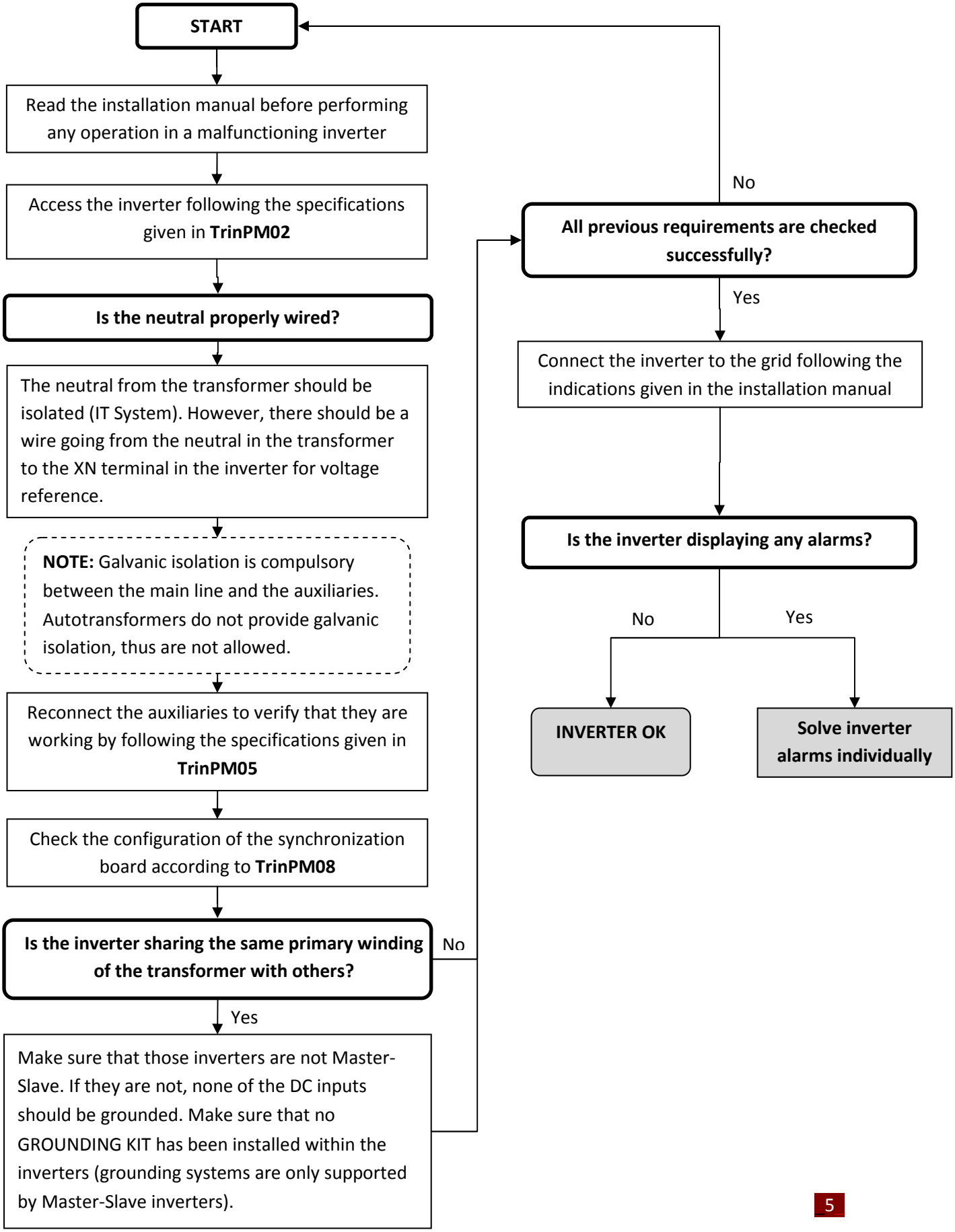
Instantaneous alarm: You will need to check the historical data from the inverter to find it because the alarm disappears when the conditions are satisfactory again.

2. AVAILABLE MEDIA:

Trin	Title
TrinPM01	Service general concepts
TrinPM02	Access to the inverter
TrinPM03	Electronic block replacement (Hardware)
TrinPM04	Electronic block replacement (Software)
TrinPM05	Auxiliary power verification
TrinPM06	Insulation failure checklist
TrinPM07	Display card verification and replacement
TrinPM08	Synchronization between inverters
TrinPM09	IGBT/Bus capacitor torque verification
TrinPM10	AC & DC surge arrestor replacement
TrinPM11	DC fuse replacement
TrinPM12	AC fuse replacement
TrinPM13	Ventilation test and fan replacement
TrinPM14	NTC and SW verification
TrinPM15	Measuring board relays replacement
TrinPM16	Varistor replacement
TrinPM17	Communication with inverters through RS485
TrinPM18	Scope
TrinPM19	Firmware upgrade
TrinPM20	Inverter calibration
TrinPM21	Inverter reconnection
TrinPM23	Wiring continuity regarding 0040 alarm
TrinPM24	Voltage measurement regarding 0040 alarm
TrinPM25	Wiring continuity regarding 0100 alarm
TrinPM26	Voltage measurement regarding 100 alarm
TrinPM27	Impedance measurement of inverter DC side
TrinPM28	Impedance measurement of inverter auxiliaries
TrinPM29	Impedance measurement of inverter AC side

Videos can be found at: **www.ingeconsuntraining.com**

3. INITIAL REQUIREMENTS



0001/0002/0003 Alarm

Grid frequency/voltage out of range

Stop the inverter by pressing the emergency push button or the display (the 1000 alarm will pop up) and check the AC Voltages (Vac1 Vac2 Vac3) at the display for every node

NOTE: If AC Voltages are zero, check the state of the AC thermal magnetic breaker (at the inverter or elsewhere in the shelter)

Access the inverter following the specifications given in **TrinPM02** and reconnect the AC power and the auxiliaries

Measure the AC voltages at the inverter AC output terminal bars. If these values coincide with the readings in the display, confirm that they are within the limits of the inverter configuration using INGECON SUN Manager.

Are the measured AC voltages within the AC range of the inverter?

No

Grid conditions are not satisfactory for the inverter

Check the state of any other components in your PV plant apart from the inverters and share with the utility the incident detected if needed.

Yes

Focus your attention on each node displaying the alarm

Disconnect the AC power and the auxiliaries. Confirm that none of the following AC fuses are tripped (measuring continuity or impedance): **FAC(n).1**, **FAC(n).2** and **FAC(n).3**. Replace any tripped AC fuse following the indications given in **TrinPM12** and check the status of the power block before reconnecting the inverter (follow indications in **E-Block verification** (AAV2011IKH04)).

OK

NOT OK

Turn on the **Q4** thermal magnetic breaker in case it is tripped.

OK

NOT OK

Disconnect the AC power and the auxiliaries. Check the state of the varistors located in the electronic block following the installation manual instructions. Replace any damaged varistor following the indications given in **TrinPM16**

OK

NOT OK

INVERTER OK

Open the SCOPE tool in INGECON SUN Manager following the indications given in **TrinPM18**. Plot Vac1 Vac2 Vac3 and contact INGETEAM Technical support if they do not present an adequate waveform.

NOT OK

Measure the AC voltage at the inverter AC output terminal bars using a multimeter. Compare these values with the ones shown in the display. Calibrate the inverter if these values do not coincide following the indications given in **TrinPM20**.

OK

NOT OK

Check the state of the AC surge arresters. Replace any damaged AC surge arrester following the indications given in **TrinPM10.**

OK

0004 Alarm — **Control current saturation**

Check in the display what electronic blocks present this alarm and access the inverter following the specifications given in **TrinPM02**

Review the wiring of the current sensor located at the AC output bars from the affected electronic blocks and solve any incidences if needed.

OK → **INVERTER OK**

NOT OK ↓

Can you access to the back inductors of the inverter through its back panel?

Yes → Visually check if the back inductors are damaged and contact INGETEAM SAT if they present any anomaly.

OK → **INVERTER OK**

NOT OK ↓

No → Disconnect Q2 and measure the resistance between:

- Each of the 3 AC output phases and **Q2(n).1** - **Q2(n).3** – **Q2(n).5** respectively
- **Q2(n).1** - **Q2(n).3** – **Q2(n).5** and **K1(n).1** - **K1(n).3** - **K1(n).5** respectively

Do you always get a measured resistance below 2 Ω?

No → The back inductors are damaged and you must visually check them by removing both the electronic block and its holding plate or the lateral panels of the inverter. Contact INGETEAM SAT if they present any anomaly.

Yes ↓

Open the SCOPE tool in INGECON SUN Manager following the indications given in **TrinPM18**. Check the **Debug Desc** values.

Debug Desc 40? → Debug Desc 92? → Debug Desc 107,108, 109?

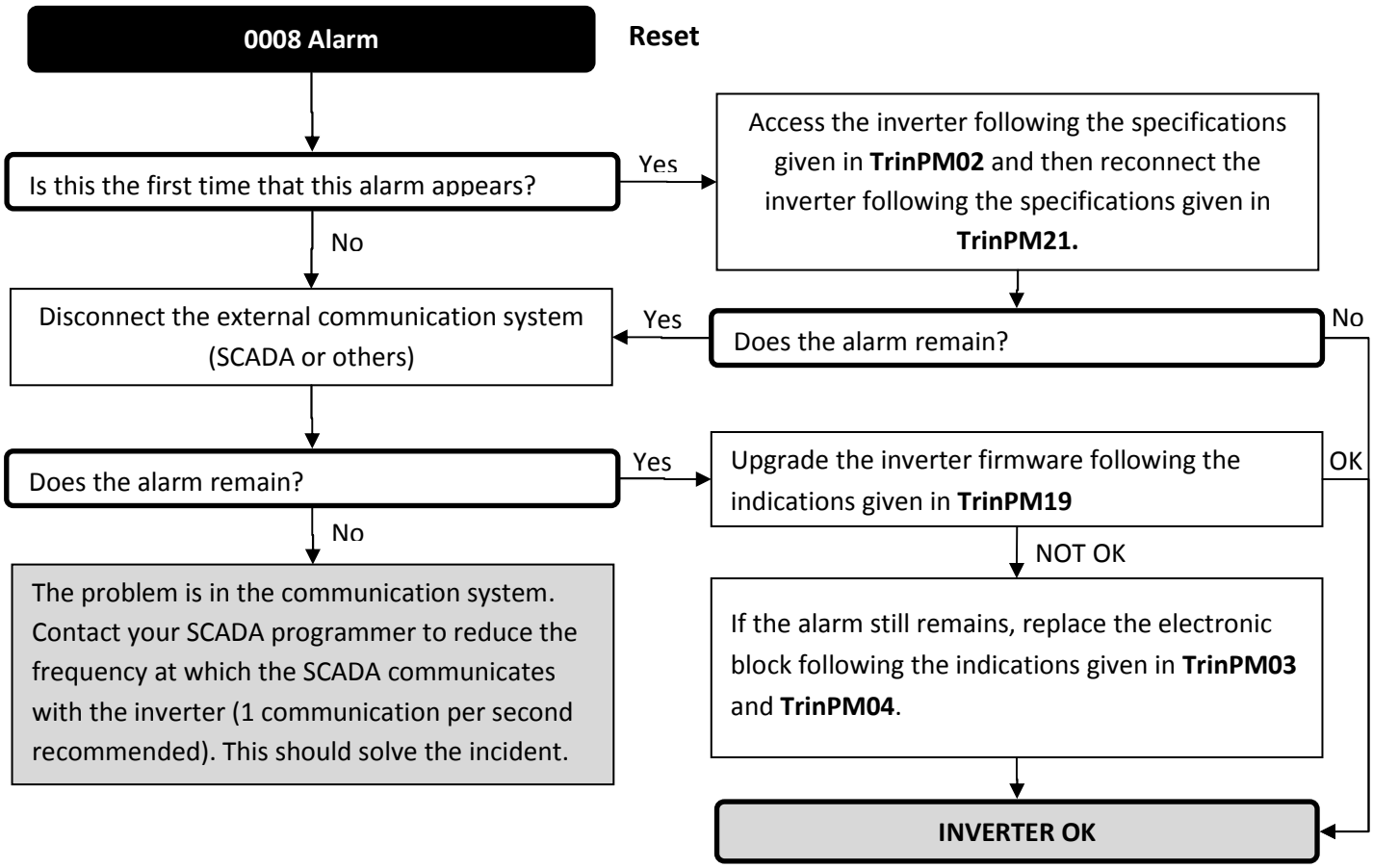
Debug Desc 40?: Check that the J4, J5, J21 connectors are well connected to the measuring board.

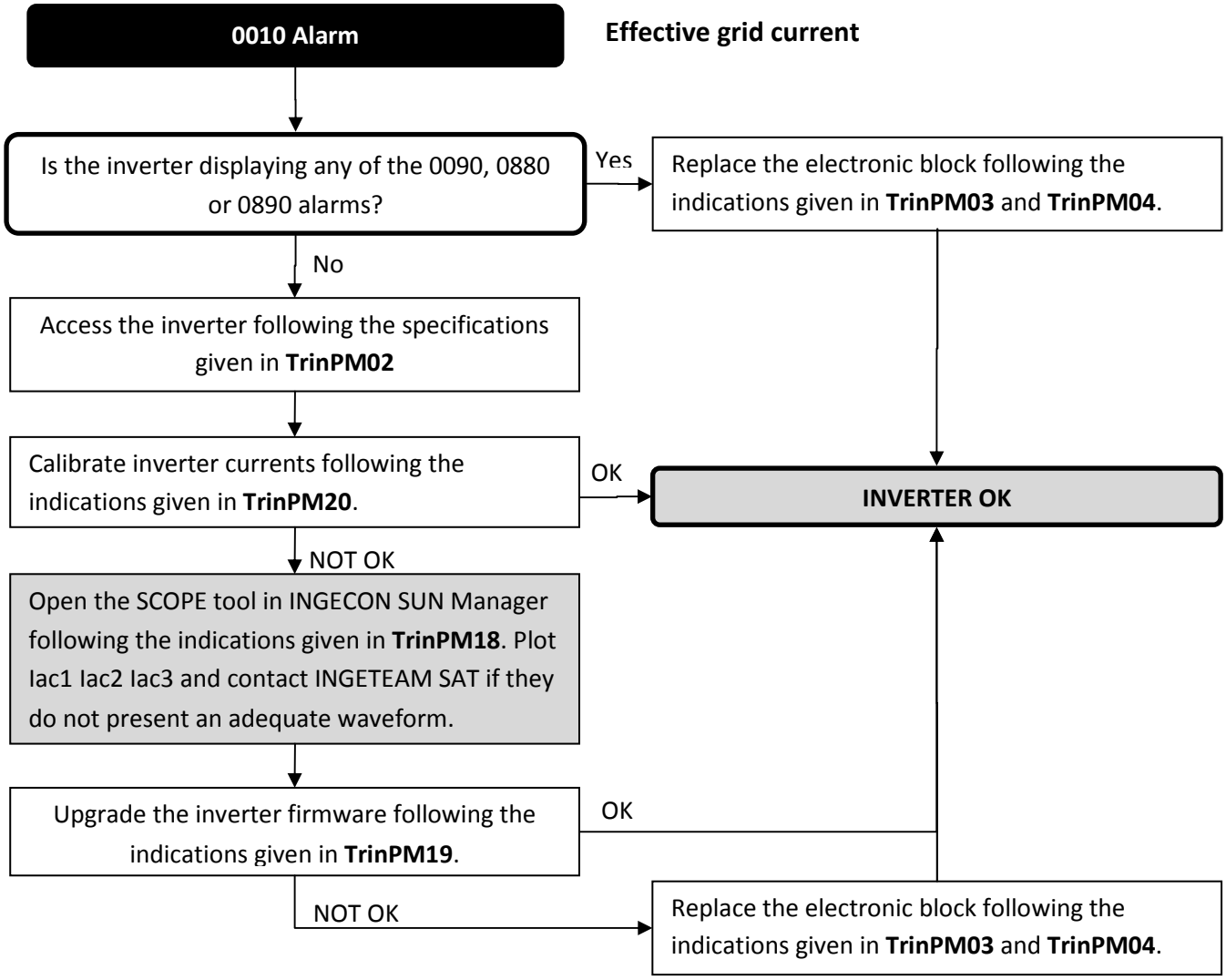
Debug Desc 92?: Calibrate the reactive power following the indications given in **TrinPM20.**

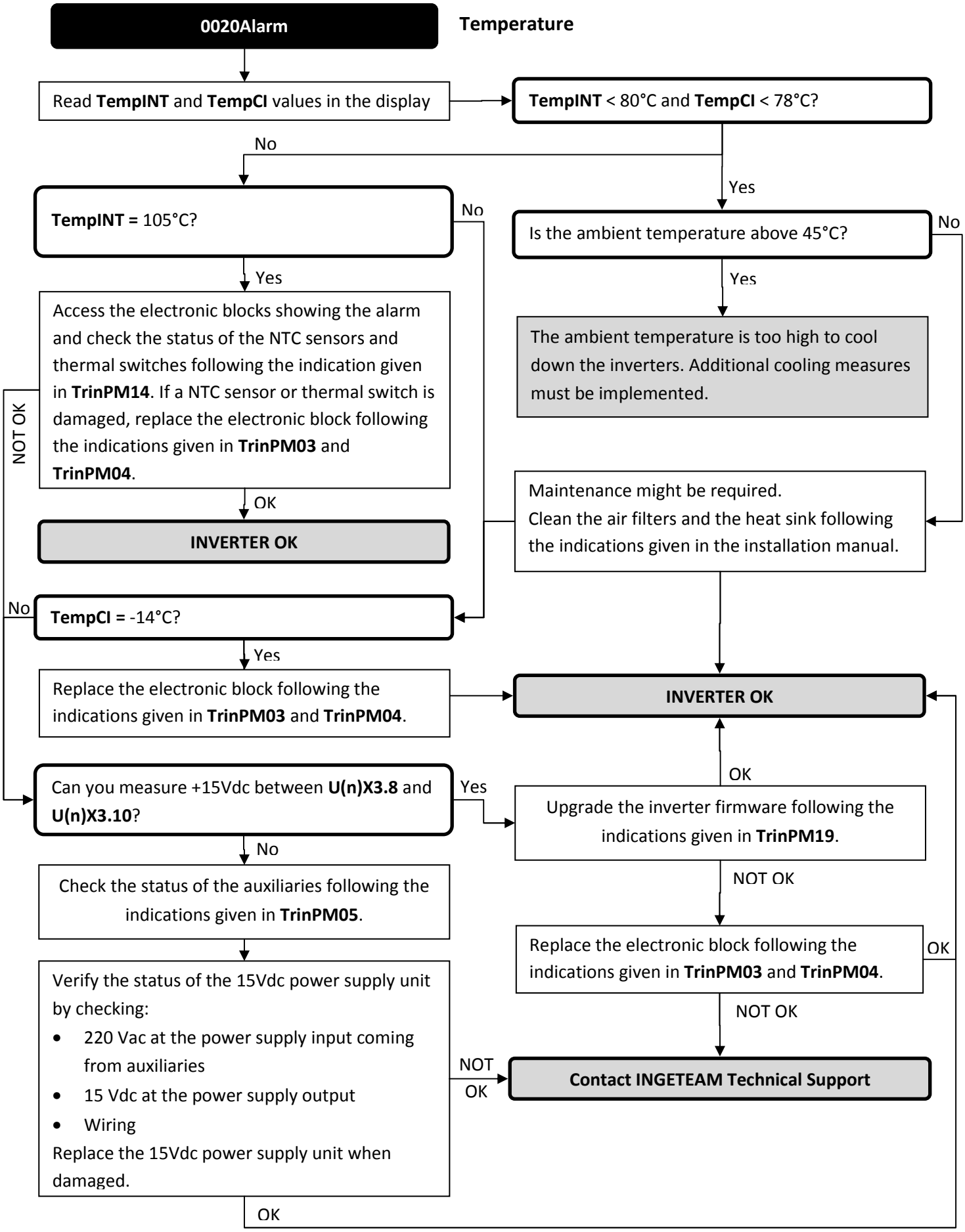
Debug Desc 107,108, 109?: Check Vdc values in the display → **Go to alarm 8000**

Any other Debug Desc? → If the alarm still remains, replace the electronic block following the indications given in **TrinPM03** and **TrinPM04**.

INVERTER OK or
Contact INGETEAM Technical Support


0008 Alarm
Reset
Is this the first time that this alarm appears?
Yes
Access the inverter following the specifications given in TrinPM02 and then reconnect the inverter following the specifications given in TrinPM21.
No
Disconnect the external communication system (SCADA or others)
Yes
Does the alarm remain?
No
Does the alarm remain?
Yes
Upgrade the inverter firmware following the indications given in TrinPM19
OK
No
NOT OK
The problem is in the communication system. Contact your SCADA programmer to reduce the frequency at which the SCADA communicates with the inverter (1 communication per second recommended). This should solve the incident.
If the alarm still remains, replace the electronic block following the indications given in TrinPM03 and TrinPM04.
INVERTER OK


0010 Alarm
Effective grid current
Is the inverter displaying any of the 0090, 0880 or 0890 alarms?
Yes
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.
No
Access the inverter following the specifications given in TrinPM02
Calibrate inverter currents following the indications given in TrinPM20.
OK
INVERTER OK
NOT OK
Open the SCOPE tool in INGECON SUN Manager following the indications given in TrinPM18. Plot Iac1 Iac2 Iac3 and contact INGETEAM SAT if they do not present an adequate waveform.
Upgrade the inverter firmware following the indications given in TrinPM19.
OK
NOT OK
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.


0020Alarm
Temperature
Read TempINT and TempCI values in the display
TempINT < 80°C and TempCI < 78°C?
No
Yes
TempINT = 105°C?
No
Yes
Is the ambient temperature above 45°C?
No
Yes
Access the electronic blocks showing the alarm and check the status of the NTC sensors and thermal switches following the indication given in TrinPM14. If a NTC sensor or thermal switch is damaged, replace the electronic block following the indications given in TrinPM03 and TrinPM04.
NOT OK
OK
The ambient temperature is too high to cool down the inverters. Additional cooling measures must be implemented.
INVERTER OK
Maintenance might be required.
Clean the air filters and the heat sink following the indications given in the installation manual.
No
TempCI = -14°C?
Yes
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.
INVERTER OK
OK
Can you measure +15Vdc between U(n)X3.8 and U(n)X3.10?
Yes
Upgrade the inverter firmware following the indications given in TrinPM19.
No
Check the status of the auxiliaries following the indications given in TrinPM05.
NOT OK
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.
OK
Verify the status of the 15Vdc power supply unit by checking:
• 220 Vac at the power supply input coming from auxiliaries
• 15 Vdc at the power supply output
• Wiring
Replace the 15Vdc power supply unit when damaged.
NOT OK
NOT OK
Contact INGETEAM Technical Support
OK

0040Alarm **Hardware fault**

Does the inverter have a LVRT kit?

Yes → If Vdc < 400 V, the 0040 alarm will appear. Wait for Vdc to reach higher voltage levels.

No ↓

Access the inverter following the specifications given in **TrinPM02**

→ Extract the fiber optic cables from the affected electronic block and reconnect the auxiliaries. Check the state of the fiber optic cables and the synchronism card following the indications given in **TrinPM08**.

Can you see the red light at the end of the fiber optic cables?

Yes → Check the wiring by following the indications given in **TrinPM23**

No ↓

Can you measure 220V AC between J1.1 and J1.2 terminals at the synchronism card?

Yes → Contact INGETEAM to get a replacement of the synchronism card / figer optic cables.

No ↓

Check the wiring by confirming:

- 220Vac between X8.4 and X8.8.
- Continuity between X8.4 and the J1.1 connector at the synchronism card
- Continuity between X8.8 and the J1.2 connector at the synchronism card

→ **INVERTER OK or Contact INGETEAM Technical Support**

Check the wiring by following the indications given in **TrinPM23**

NOT OK → Open the SCOPE tool in INGECON SUN Manager following the indications given in **TrinPM18**. Check the **Debug Desc** values.

OK ↓

INVERTER OK

Debug Desc 55 or 56?

No → Debug Desc 119?

Yes → In Master-Slave configurations, when the master stops, the rest of slaves do also stop. Reconnect the inverter to solve the problem.

Debug Desc 119?

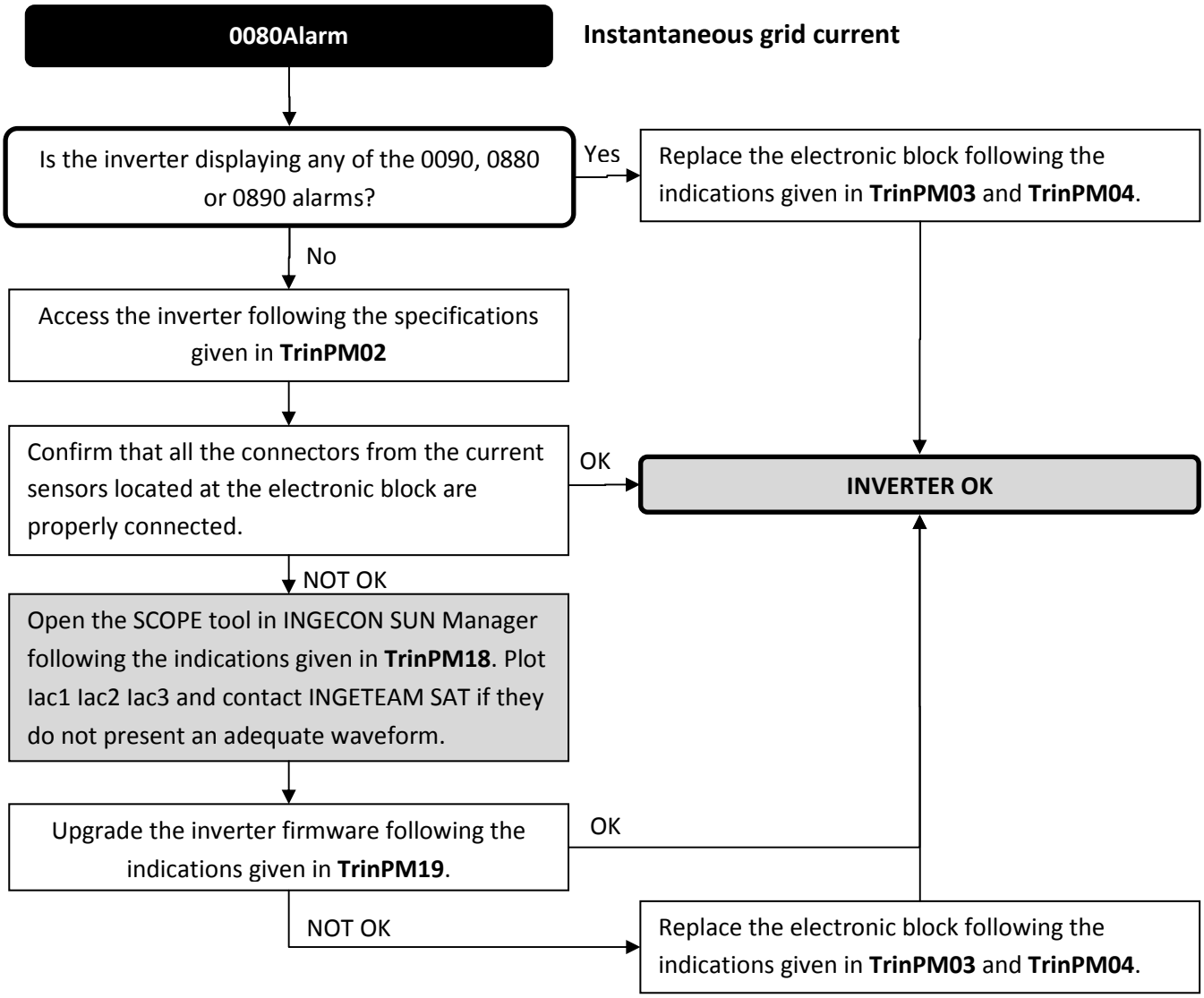
No → **INVERTER OK or Contact INGETEAM Technical Support**

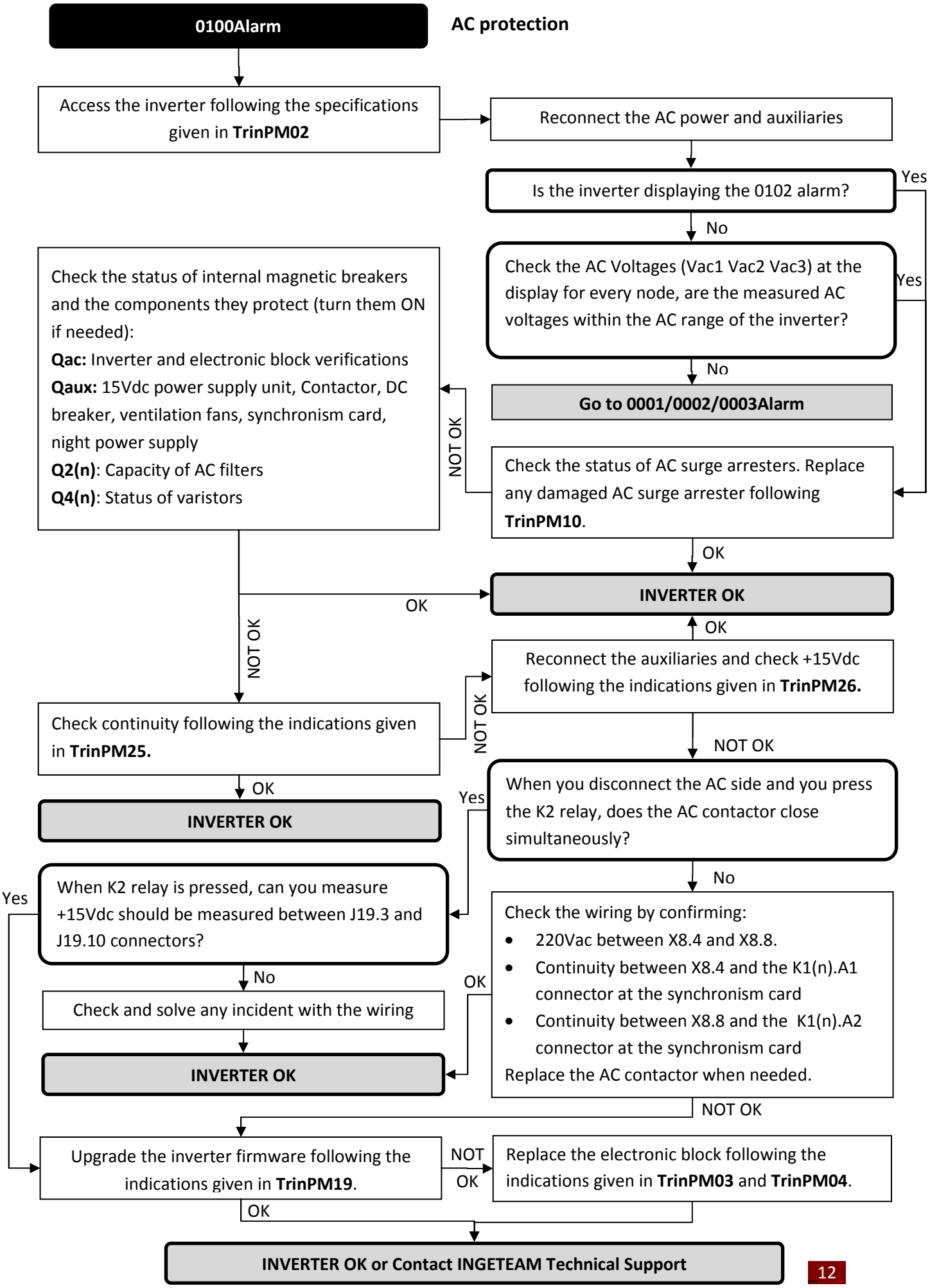
Yes ↓

Check the status of the DC contactor or the DC switch. Check the status of the LVRT kit.

↓

INVERTER OK or Contact INGETEAM Technical Support


0080Alarm
Instantaneous grid current
Is the inverter displaying any of the 0090, 0880 or 0890 alarms?
Yes
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.
No
Access the inverter following the specifications given in TrinPM02
Confirm that all the connectors from the current sensors located at the electronic block are properly connected.
OK
INVERTER OK
NOT OK
Open the SCOPE tool in INGECON SUN Manager following the indications given in TrinPM18. Plot Iac1 Iac2 Iac3 and contact INGETEAM SAT if they do not present an adequate waveform.
Upgrade the inverter firmware following the indications given in TrinPM19.
OK
NOT OK
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.


0100Alarm
AC protection
Access the inverter following the specifications given in TrinPM02
Reconnect the AC power and auxiliaries
Is the inverter displaying the 0102 alarm?
Yes
No
Check the AC Voltages (Vac1 Vac2 Vac3) at the display for every node, are the measured AC voltages within the AC range of the inverter?
Yes
No
Go to 0001/0002/0003Alarm
Check the status of internal magnetic breakers and the components they protect (turn them ON if needed):
Qac: Inverter and electronic block verifications
Qaux: 15Vdc power supply unit, Contactor, DC breaker, ventilation fans, synchronism card, night power supply
Q2(n): Capacity of AC filters
Q4(n): Status of varistors
NOT OK
Check the status of AC surge arresters. Replace any damaged AC surge arrester following TrinPM10.
OK
INVERTER OK
OK
NOT OK
OK
Reconnect the auxiliaries and check +15Vdc following the indications given in TrinPM26.
NOT OK
Check continuity following the indications given in TrinPM25.
NOT OK
OK
INVERTER OK
Yes
When you disconnect the AC side and you press the K2 relay, does the AC contactor close simultaneously?
No
Yes
When K2 relay is pressed, can you measure +15Vdc should be measured between J19.3 and J19.10 connectors?
No
Check and solve any incident with the wiring
INVERTER OK
OK
Check the wiring by confirming:
• 220Vac between X8.4 and X8.8.
• Continuity between X8.4 and the K1(n).A1 connector at the synchronism card
• Continuity between X8.8 and the K1(n).A2 connector at the synchronism card
Replace the AC contactor when needed.
NOT OK
Upgrade the inverter firmware following the indications given in TrinPM19.
NOT OK
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.
OK
INVERTER OK or Contact INGETEAM Technical Support

0200Alarm **DC protection**

Access the inverter following the specifications given in **TrinPM02**

Check the state of the DC surge arresters. Replace any damaged DC surge arrester following the indications given in **TrinPM10.**

OK

NOT OK

Can continuity be measured between the following points?

- IRVDC.n.11 – IRVDC.n.12
- XMDC(n+2) – U(n).J19.1

Yes

Do you get a large resistance when you measure the impedance between the following points?

- IRVDC.n.11 – Ground
- IRVDC.n.12 – Ground

Yes

No

No

Review wiring.

OK

INVERTER OK or Contact INGETEAM Technical Support

OK

Review and replace any damaged fused if you do not get continuity between **XFDC.1 – XFDC.2**

Yes

Does the inverter have a *"blown fuse sensor at the DC input"* kit?

NOT OK

Does the inverter have a grounding kit?

No

No

Yes

Reconnect the auxiliaries and confirm 15Vdc between the following points (solve any wiring incident if you not detect 15 Vd):

- **XFDC.1 – X7DC.4.**
- **XMDC.1 - X7DC.4.**
- **XMON(n).n – XDIS(n).0.**
- **U(n)X3.1 - U(n)X3.10.**
- **U(n)J19.1 - U(n)J19.10.**

Turn on (or replace) the thermal magnetic breaker included in the grounding kit if you do not get continuity between **XFDC.2 – XFDC.3**

Verify the status of the 15Vdc power supply unit by checking:

- 220 Vac at the power supply input coming from auxiliaries
- 15 Vdc at the power supply output
- Wiring

Replace the 15Vdc power supply unit when damaged.

Do you measure 15 Vdc between all those points?

No

Yes

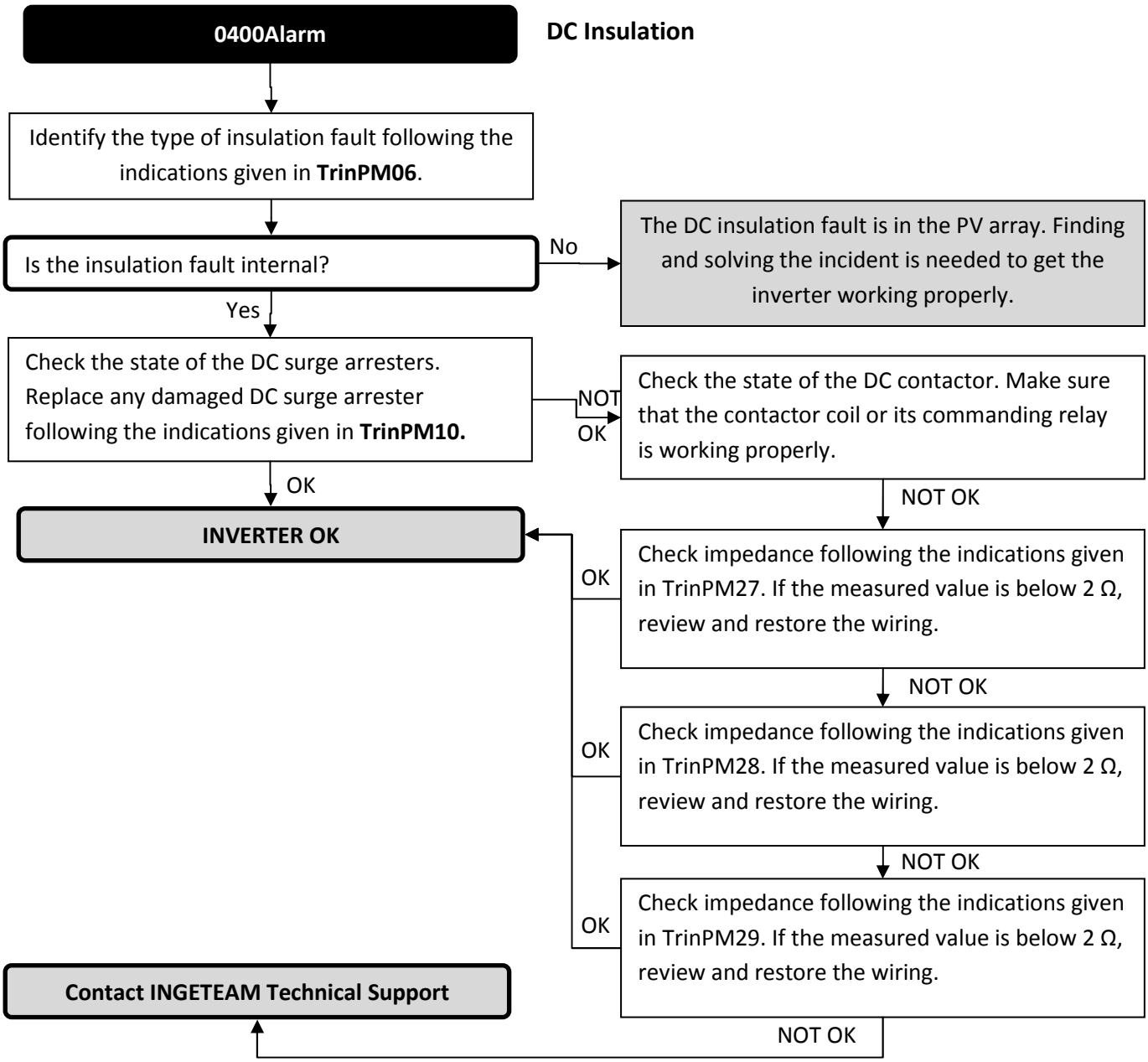
Upgrade the inverter firmware following the indications given in **TrinPM19**.

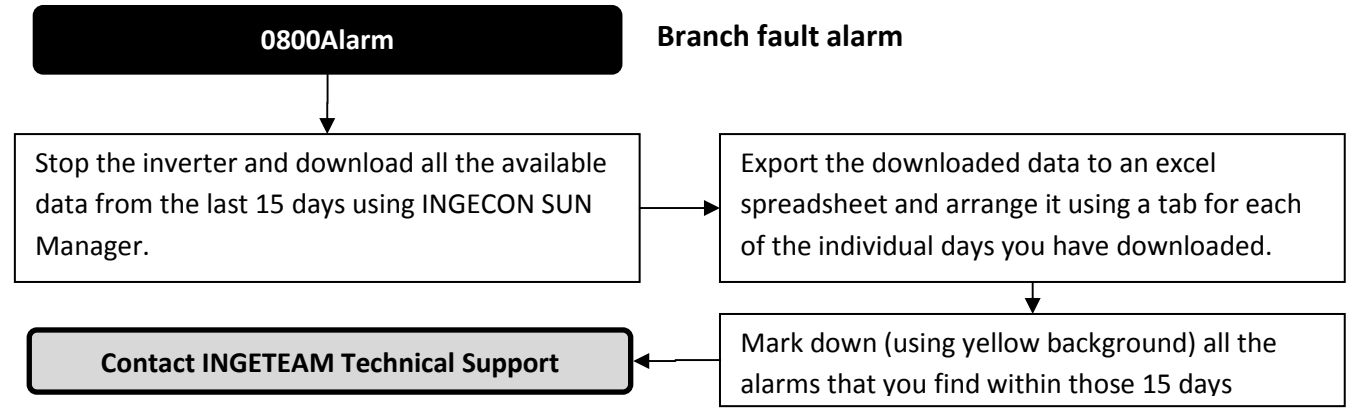
NOT OK

Replace the electronic block following the indications given in **TrinPM03** and **TrinPM04**.

OK

INVERTER OK or Contact INGETEAM Technical Support


0400Alarm
DC Insulation
Identify the type of insulation fault following the indications given in TrinPM06.
Is the insulation fault internal?
No
The DC insulation fault is in the PV array. Finding and solving the incident is needed to get the inverter working properly.
Yes
Check the state of the DC surge arresters. Replace any damaged DC surge arrester following the indications given in TrinPM10.
NOT OK
Check the state of the DC contactor. Make sure that the contactor coil or its commanding relay is working properly.
OK
INVERTER OK
NOT OK
OK
Check impedance following the indications given in TrinPM27. If the measured value is below 2 Ω, review and restore the wiring.
NOT OK
OK
Check impedance following the indications given in TrinPM28. If the measured value is below 2 Ω, review and restore the wiring.
NOT OK
OK
Check impedance following the indications given in TrinPM29. If the measured value is below 2 Ω, review and restore the wiring.
Contact INGETEAM Technical Support
NOT OK


0800Alarm
Branch fault alarm
Stop the inverter and download all the available data from the last 15 days using INGECON SUN Manager.
Export the downloaded data to an excel spreadsheet and arrange it using a tab for each of the individual days you have downloaded.
Mark down (using yellow background) all the alarms that you find within those 15 days
Contact INGETEAM Technical Support

1000Alarm

Manual shutdown

Is the inverter displaying the 1000 alarm in combination with other alarm codes?

No → Have you checked the following points:

- The emergency button is released
- The inverter is not stopped from the display
- Every inverter door is properly closed

NOT OK → Access the inverter following the specifications given in **TrinPM02**

OK → **INVERTER OK**

Yes → Access the inverter following the specifications given in **TrinPM02** and reconnect auxiliaries.

Is the inverter displaying the 1320 alarm?

No → Is the inverter displaying any of the 1360 or 1363 alarms?

Yes → Confirm that the +15 Vdc power supply unit LED is ON. Replace it in case it is damaged.

OK → **INVERTER OK**

NOT OK → **Review the other alarms individually**

Check the state of the auxiliaries.

OK → **INVERTER OK**

NOT OK → **Review the other alarms individually**

Yes → Is the inverter displaying any of the 1360 or 1363 alarms?

No → **Review the list of alarms individually**

Yes → Do you get continuity between the following points?

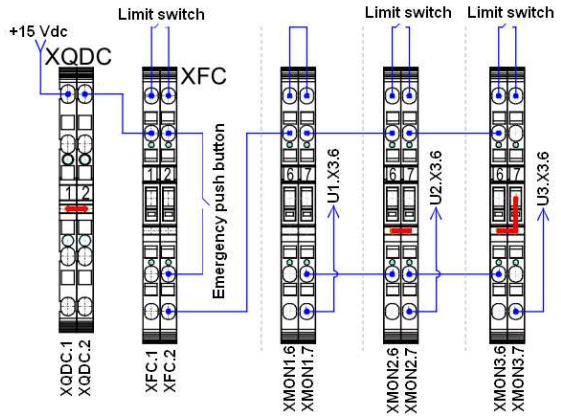
- XMON(n).7 - U(n)X3.6
- U(n)X3.6 - U(n)J19.8

Yes → Reconnect auxiliaries. Do you get +15Vdc between the following points?

- U(n)J19.8 - U(n)J19.10

No → Review wiring

No → Manually disable every limit switch and release the emergency push button. Check continuity:



*In the case of symmetric inverter, check with INGETEAM Technical Support to get this sequence.

OK → **INVERTER OK or Contact INGETEAM Technical Support**

No → Was the inverter stopped by communications?

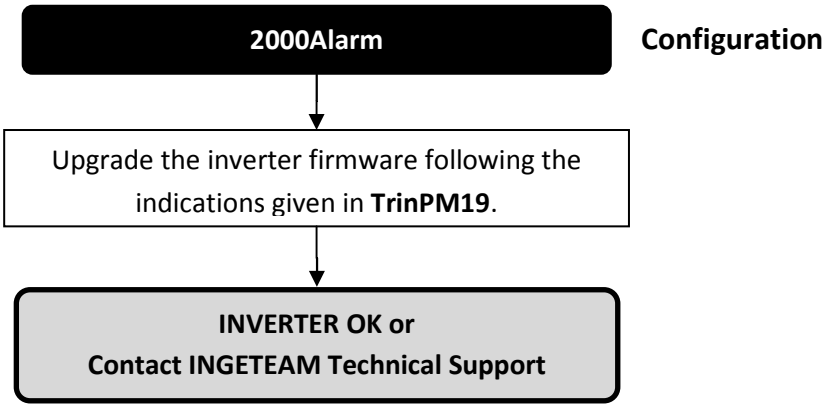
Yes → The problem is in the communication system. Contact your SCADA programmer to solve the incident

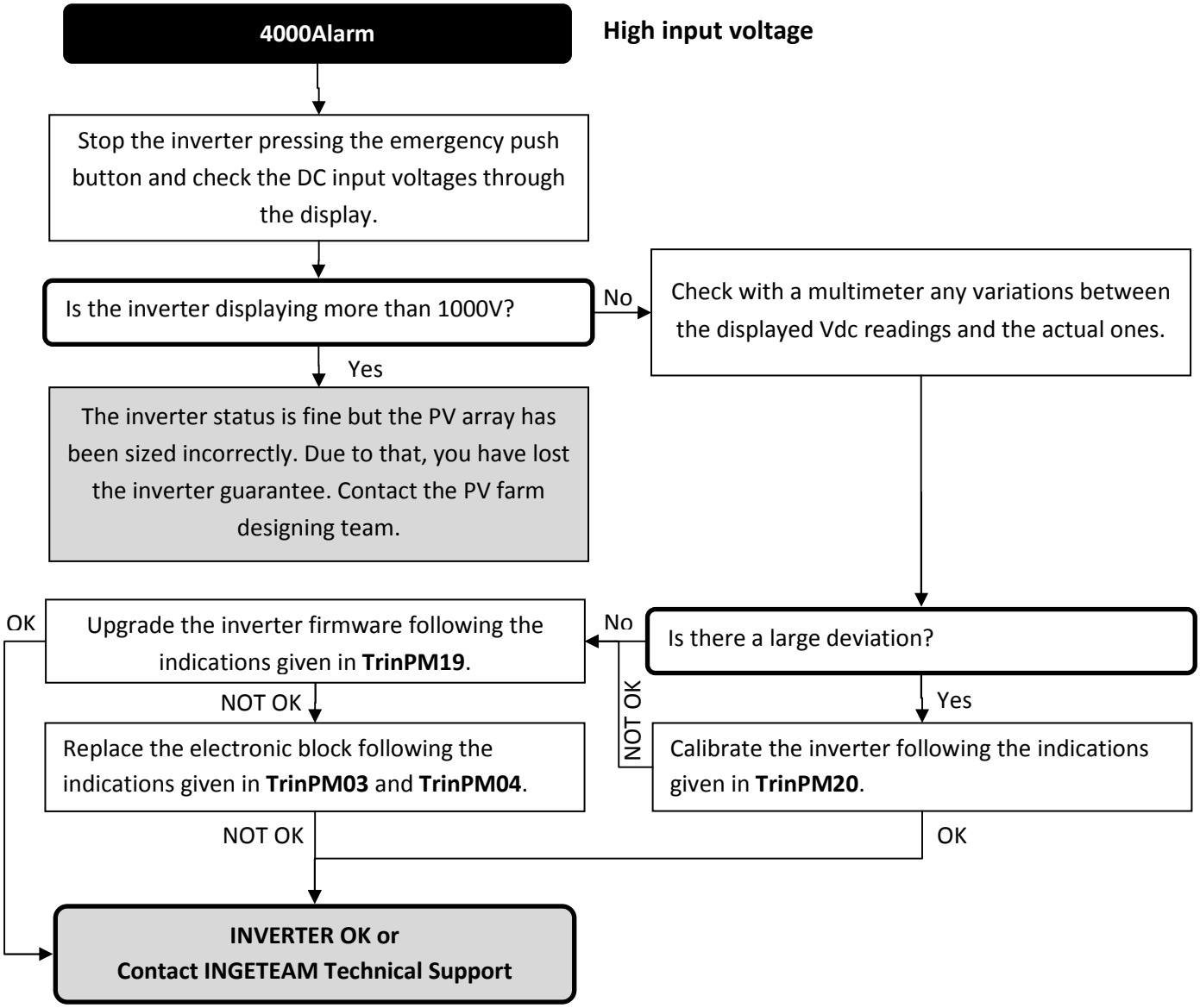
Yes → Upgrade the inverter firmware following the indications given in **TrinPM19**.

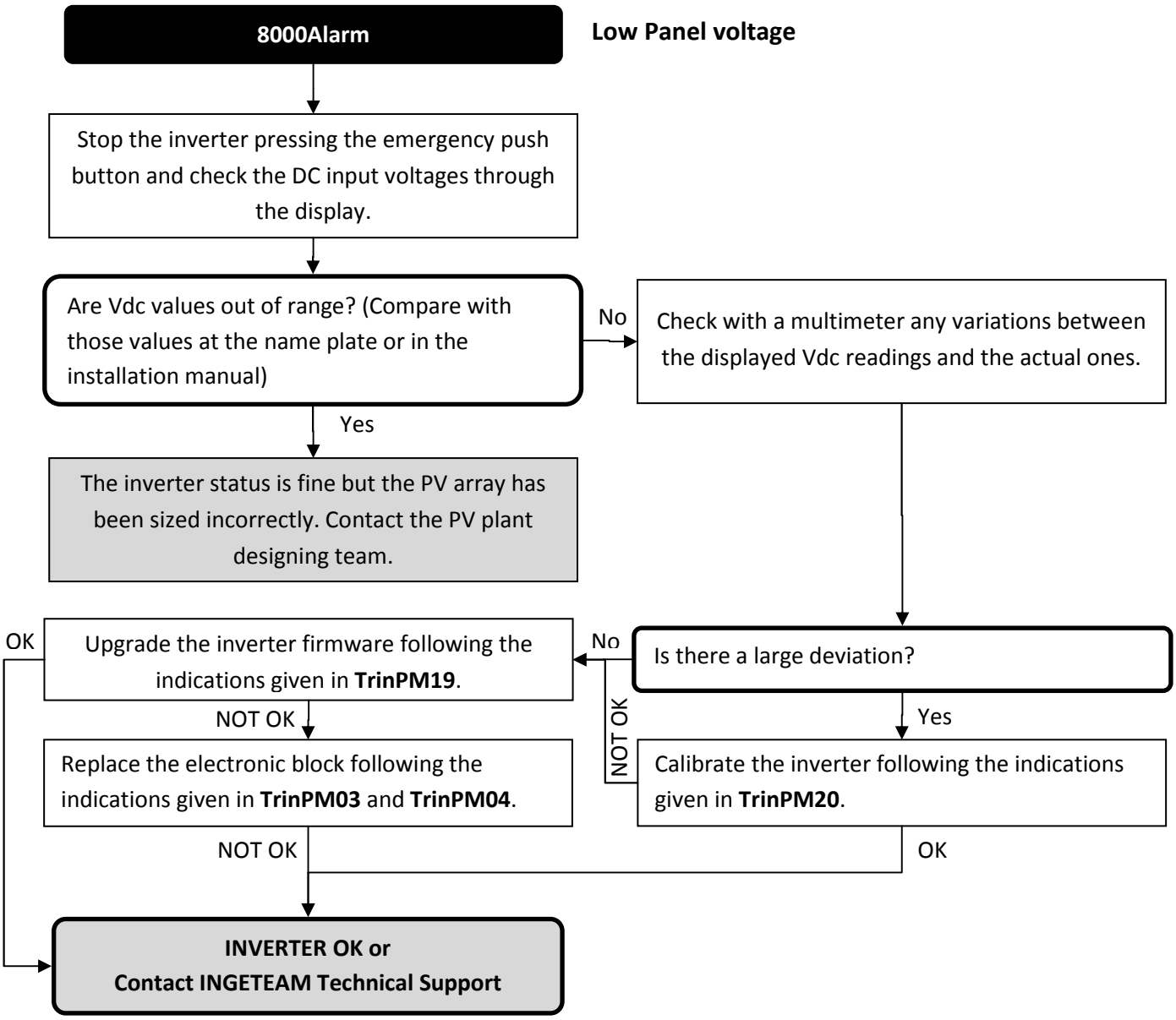
NOT OK → Manually disable every limit switch and release the emergency push button. Check continuity:

OK → **INVERTER OK or Contact INGETEAM Technical Support**

NOT OK → Replace the electronic block following the indications given in **TrinPM03** and **TrinPM04**.


2000Alarm
Configuration
Upgrade the inverter firmware following the indications given in **TrinPM19**.
INVERTER OK or
Contact INGETEAM Technical Support


4000Alarm
High input voltage
Stop the inverter pressing the emergency push button and check the DC input voltages through the display.
Is the inverter displaying more than 1000V?
No
Check with a multimeter any variations between the displayed Vdc readings and the actual ones.
Yes
The inverter status is fine but the PV array has been sized incorrectly. Due to that, you have lost the inverter guarantee. Contact the PV farm designing team.
OK
Upgrade the inverter firmware following the indications given in **TrinPM19**.
No
Is there a large deviation?
NOT OK
Yes
NOT OK
Replace the electronic block following the indications given in **TrinPM03** and **TrinPM04**.
Calibrate the inverter following the indications given in **TrinPM20**.
NOT OK
OK
INVERTER OK or
Contact INGETEAM Technical Support


8000Alarm
Low Panel voltage
Stop the inverter pressing the emergency push button and check the DC input voltages through the display.
Are Vdc values out of range? (Compare with those values at the name plate or in the installation manual)
No
Check with a multimeter any variations between the displayed Vdc readings and the actual ones.
Yes
The inverter status is fine but the PV array has been sized incorrectly. Contact the PV plant designing team.
OK
Upgrade the inverter firmware following the indications given in **TrinPM19**.
No
Is there a large deviation?
NOT OK
NOT OK
Yes
Replace the electronic block following the indications given in **TrinPM03** and **TrinPM04**.
Calibrate the inverter following the indications given in **TrinPM20**.
NOT OK
OK
INVERTER OK or
Contact INGETEAM Technical Support

7FFFH Alarm

Fatal error

NOTE: When a FATAL ERROR occurs, the inverter is unblocked by entering a code through the display.

Access the inverter following the specifications given in **TrinPM02**

- Confirm the proper connection of the wiring between the drivers' board and the IGBTS
- The 32 pin flat cable between the measuring board and the drivers' board must be properly plugged.

OK → **INVERTER OK**

NOT OK

Reconnect auxiliaries and get the DebugDesc value using INGECON SUN Manager by following the indications given in **TrinPM18**.

Do you get DebugDesc **110**?

No

Yes

Check what electronic blocks show the alarm.

Do you get two contiguous electronic blocks showing the alarm (either 1 and 2 or 3 and 4)?

No

Yes

Either the malfunctioning fan or the damaged XDFV terminal is related to the inductor´s fan located at the back of the cabinet where the alarms are shown. Substitute the XDFV terminal with one that works properly.

NOT OK

Substitute the malfunctioning fan after conducting a ventilation test following the indications given in **TrinPM13**.

INVERTER OK or Contact INGETEAM Technical Support

OK

Does the inverter have 3 electronic blocks?

No

Yes

Is the alarm in either electronic block 1 or 2?

Yes

No

Substitute in sequence every XDFV terminal associated to the third electronic block with one that works properly and if the alarm remains, substitute the malfunctioning fan after conducting a ventilation test following the indications given in **TrinPM13**.

The malfunctioning fan or damaged XDFV terminal is the one on top of that electronic block.

Substitute the XDFV terminal with one that works properly and if the alarm remains, substitute the malfunctioning fan after conducting a fan test following the indications given in **TrinPM13**.

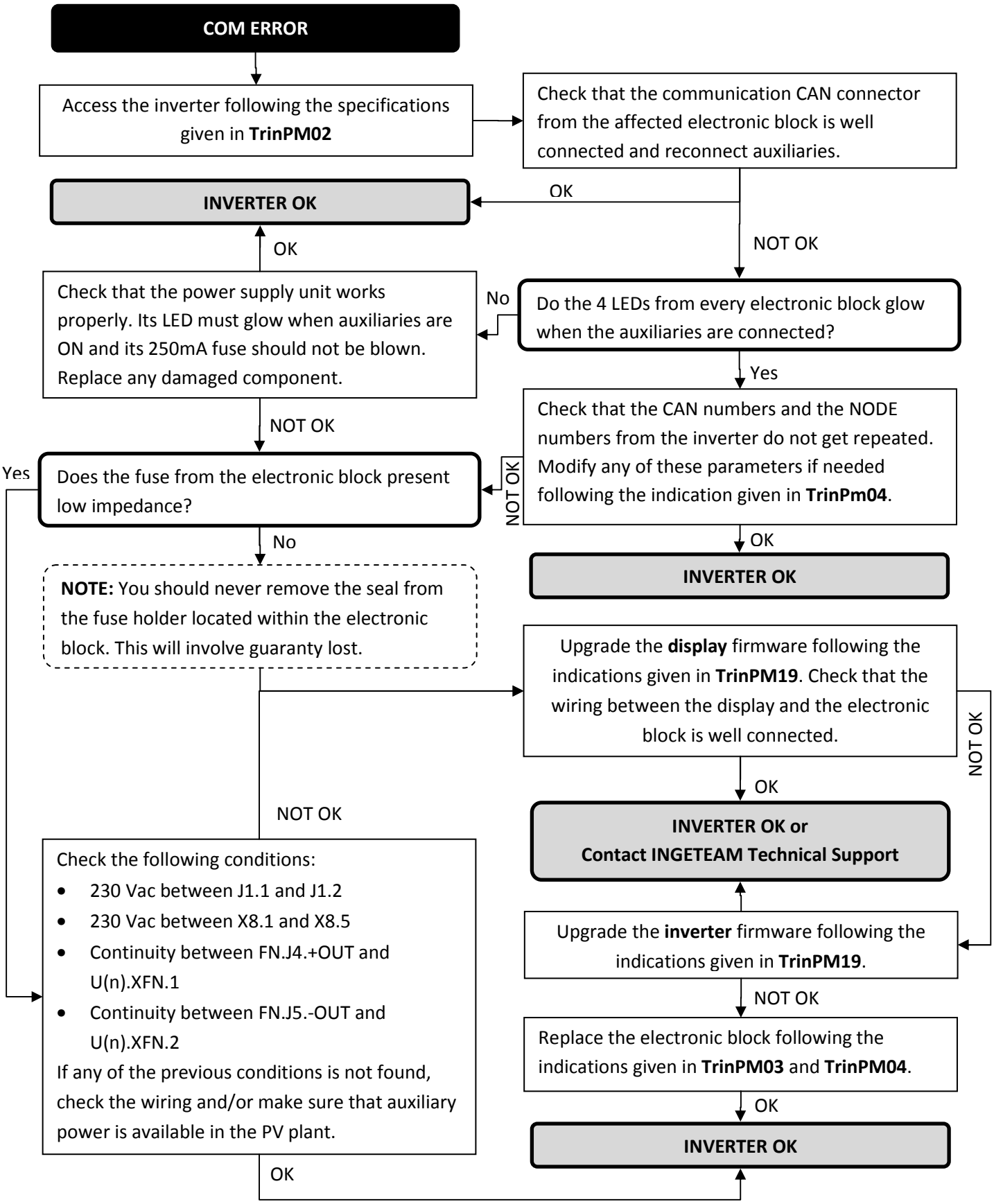
Do you get DebugDesc **21**?

No

Yes

Review the affected electronic block thermal switches by measuring continuity between **U(n)X3.8** and **U(n)J19.2**. Improve wiring if needed.

INVERTER OK or Contact INGETEAM Technical Support


COM ERROR
Access the inverter following the specifications given in TrinPM02
Check that the communication CAN connector from the affected electronic block is well connected and reconnect auxiliaries.
OK
INVERTER OK
NOT OK
OK
Check that the power supply unit works properly. Its LED must glow when auxiliaries are ON and its 250mA fuse should not be blown. Replace any damaged component.
No
Do the 4 LEDs from every electronic block glow when the auxiliaries are connected?
Yes
NOT OK
Check that the CAN numbers and the NODE numbers from the inverter do not get repeated. Modify any of these parameters if needed following the indication given in TrinPm04.
Yes
Does the fuse from the electronic block present low impedance?
NOT OK
No
OK
INVERTER OK
NOTE: You should never remove the seal from the fuse holder located within the electronic block. This will involve guaranty lost.
Upgrade the display firmware following the indications given in TrinPM19. Check that the wiring between the display and the electronic block is well connected.
NOT OK
OK
NOT OK
INVERTER OK or
Contact INGETEAM Technical Support
Check the following conditions:
230 Vac between J1.1 and J1.2
230 Vac between X8.1 and X8.5
Continuity between FN.J4.+OUT and U(n).XFN.1
Continuity between FN.J5.-OUT and U(n).XFN.2
If any of the previous conditions is not found, check the wiring and/or make sure that auxiliary power is available in the PV plant.
Upgrade the inverter firmware following the indications given in TrinPM19.
NOT OK
Replace the electronic block following the indications given in TrinPM03 and TrinPM04.
OK
INVERTER OK
OK